Deep Learning Assignment Report

Image classification on Fashion-MNIST and CIFAR-10, and study-level abnormality detection on MURA

Student: George Stamatogiannopoulos

Course: Deep Learning

Date: 1 April 2026

Executive summary

Part 1 compared an MLP and a CNN on Fashion-MNIST and CIFAR-10.
Part 2 compared a custom CNN and an EfficientNetB0 transfer-learning model on the MURA X-ray dataset.
Best Part 1 results: Fashion-MNIST MLP accuracy 0.8901; CIFAR-10 CNN accuracy 0.6795.
Best Part 2 result: EfficientNetB0 transfer with study AUC 0.8527 and study accuracy 0.7848.

Before submission: replace the placeholder code link with your GitHub / Drive / Dropbox folder URL.

Code repository / shared folder link: https://github.com/gstamatogiannopoulos/deep-learning-assignment

1. Introduction

This report documents the development and evaluation of deep learning models for two tasks. Part 1 is multi-class image classification on Fashion-MNIST and CIFAR-10. Part 2 is study-level binary classification on the MURA musculoskeletal X-ray dataset, where each study is classified as normal or abnormal.

The goal was to compare simple baselines against stronger convolutional or transfer-learning models, record practical implementation issues, and summarize what worked and what did not.

2. Experimental setup

2.1 Datasets and preprocessing

Fashion-MNIST contains grayscale 28x28 clothing images from 10 classes. CIFAR-10 contains 32x32 color images from 10 object classes. For both datasets, images were normalized to the [0, 1] range and a validation split was used during training.

For MURA, the data loader worked at image level during training but final evaluation was performed at study level. Predictions for all images in a study were averaged to produce one study probability. During preprocessing, 4 unreadable images were detected and skipped.

2.2 Models

Task	Model	Main idea	Notes
Fashion-MNIST	MLP	Flatten -> Dense layers -> Softmax	Simple fully connected baseline
Fashion-MNIST	CNN	Convolution + pooling + classifier	Light CNN; underperformed in this run
CIFAR-10	MLP	Flatten -> Dense layers -> Softmax	Weak baseline on natural images
CIFAR-10	CNN	Convolution blocks + pooling + classifier	Better fit for spatial structure
MURA	Custom CNN	CNN trained from scratch	Own model required by the assignment
MURA	EfficientNetB0	Pre-trained backbone + fine-tuning	Transfer-learning model

3. Part 1 - Image classification

3.1 Numerical results

Dataset	Model	Test accuracy	Test loss	Observation
Fashion-MNIST	MLP	0.8901	0.3171	Best result on this dataset
Fashion-MNIST	CNN	0.7524	0.6661	Weaker than expected in this run
CIFAR-10	MLP	0.4166	1.6857	Weak baseline
CIFAR-10	CNN	0.6795	0.9371	Clear improvement over MLP

Two different patterns appeared. On CIFAR-10, the CNN clearly outperformed the MLP, which is expected because CIFAR-10 contains natural images with strong local spatial structure. On Fashion-MNIST, the opposite happened in this specific run: the MLP reached higher test accuracy than the CNN.

The most likely reason is that the chosen Fashion-MNIST CNN was too small or did not train long enough to separate visually similar clothing categories.

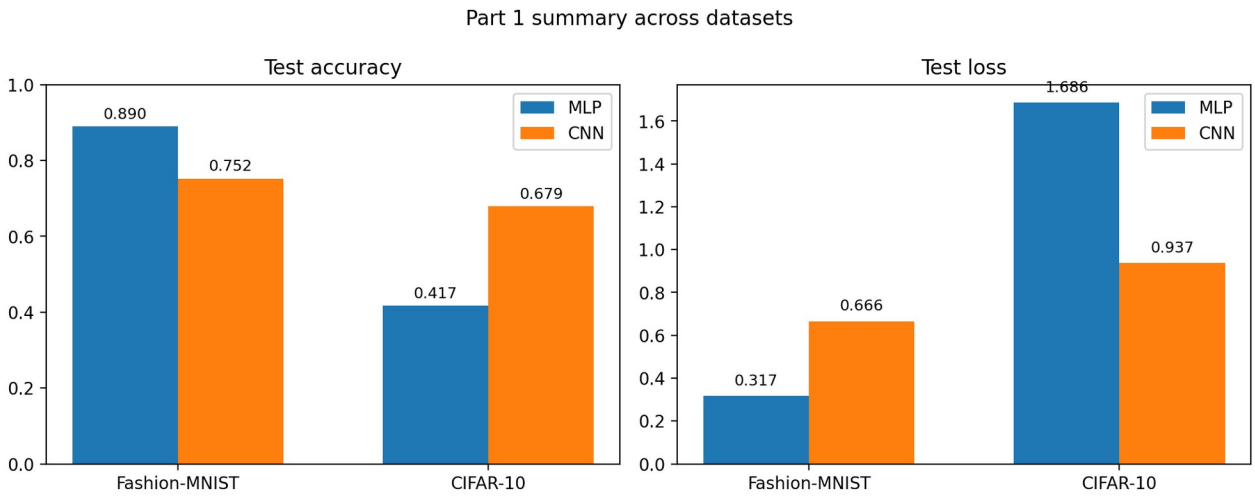

Figure 1. Part 1 summary of test accuracy and test loss.

3.2 Selected saved outputs

The next figures are selected outputs from the Part 1 scripts. They illustrate the CIFAR-10 experiment, where the CNN gave the stronger result. The confusion matrix shows strong performance on automobile, ship, truck, and frog, while larger confusions remain between some animal classes such as cat, dog, deer, and horse.

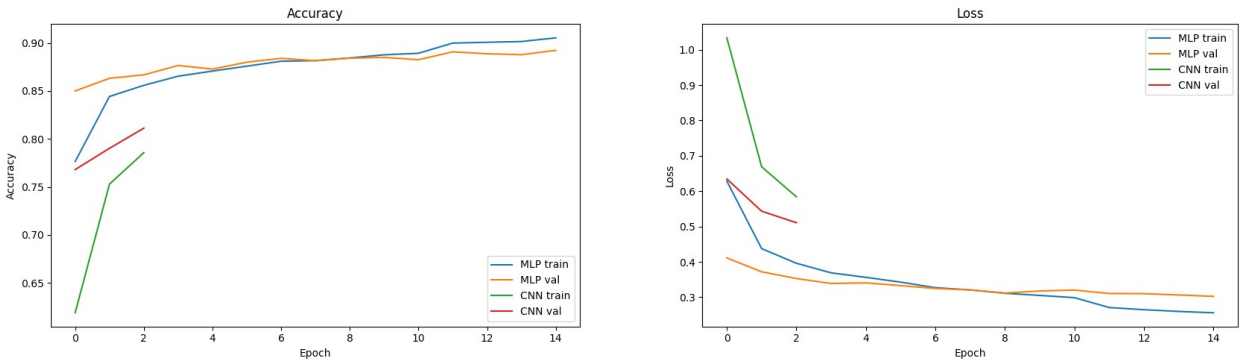

Figure 2. CIFAR-10 training accuracy and loss curves for the MLP and CNN models.

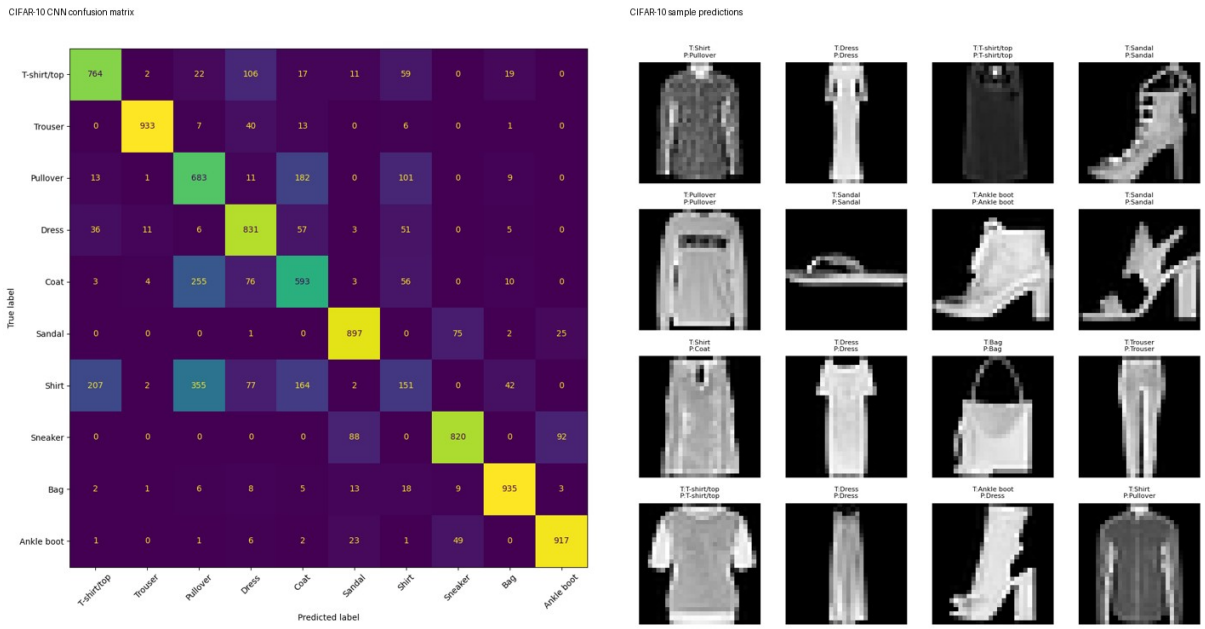

Figure 3. CIFAR-10 CNN confusion matrix and example predictions.

4. Part 2 - X-ray study classification on MURA

4.1 Training procedure

The custom CNN was trained from scratch for 8 epochs. The transfer-learning model used EfficientNetB0 with two phases: first the classification head was trained with the backbone frozen, and then the top layers were fine-tuned with a smaller learning rate.

Evaluation was performed at study level. For each study, image probabilities were averaged and then thresholded to produce a normal/abnormal decision. This makes the evaluation consistent with the assignment statement, which asks for study-level classification.

4.2 Numerical results

Model	Study AUC	Study accuracy	Study kappa	Validation studies
Custom CNN	0.7616	0.6914	0.3551	1199
EfficientNetB0 transfer	0.8527	0.7848	0.5547	1199

The transfer-learning model was better on every study-level metric. This is the expected outcome, because EfficientNetB0 starts from much stronger visual features than a small CNN trained from scratch on the target dataset only.

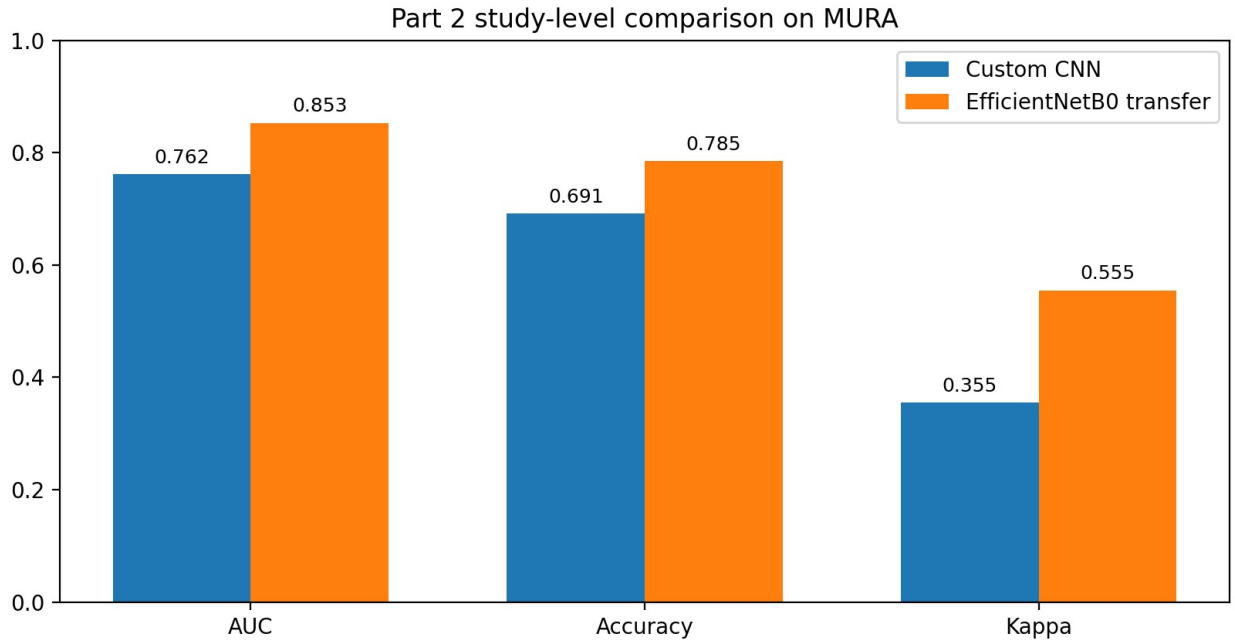

Figure 4. Study-level comparison of the two MURA models.

4.3 Diagnostic figures

The custom CNN confusion matrix shows that abnormal studies were still often predicted as normal. The transfer-learning model reduced this problem and produced a better balance between sensitivity and specificity.

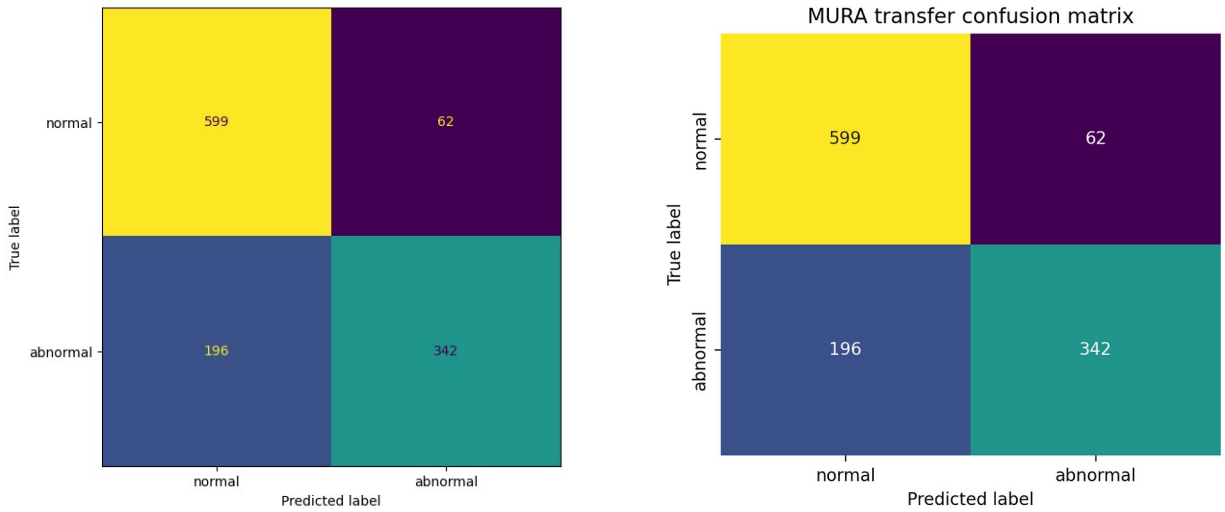

Figure 5. Study confusion matrices: custom CNN (left) and EfficientNetB0 transfer (right).

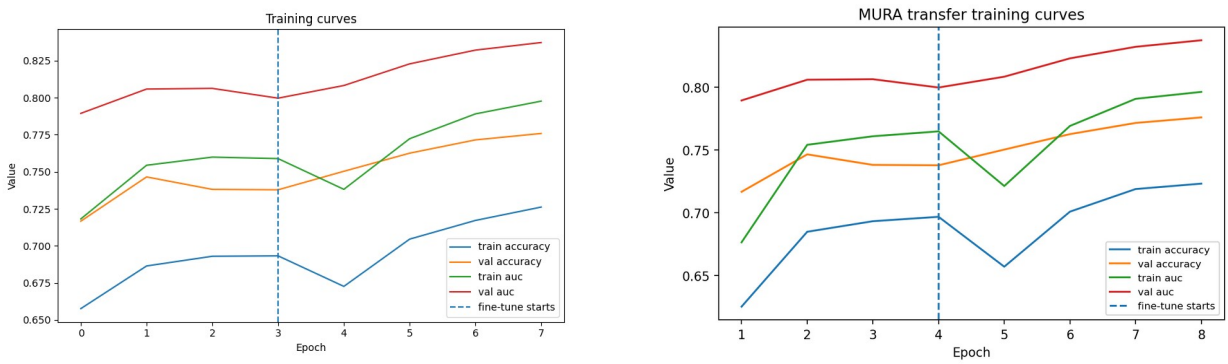

Figure 6. Training curves: custom CNN (left) and EfficientNetB0 transfer (right).

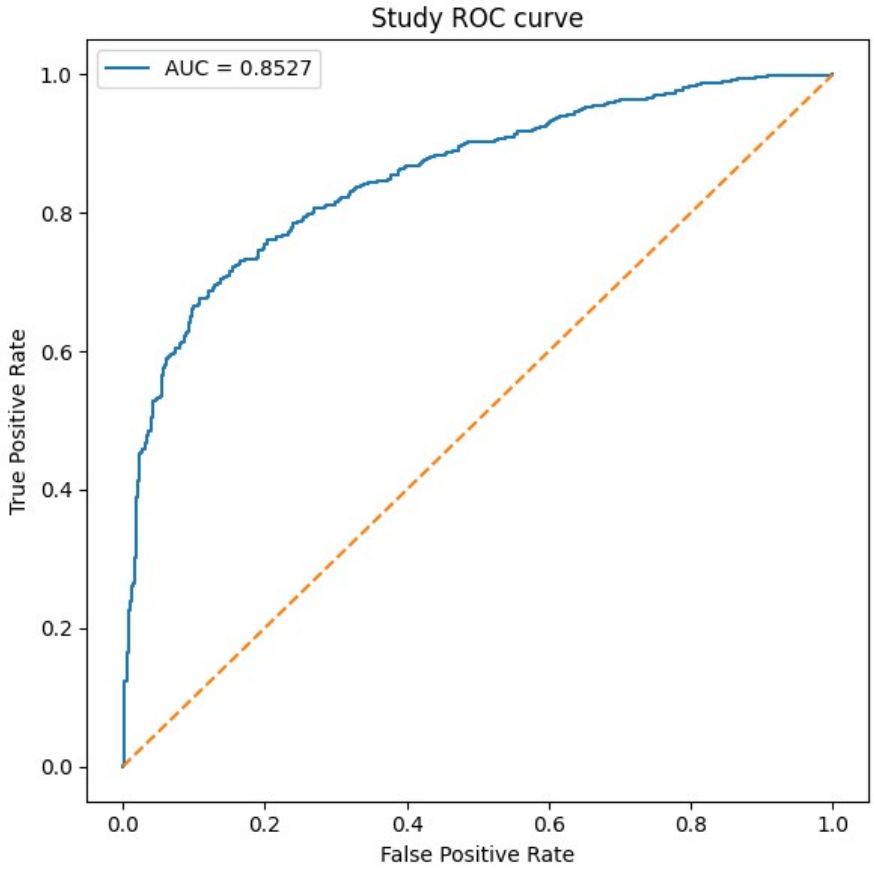

Figure 7. ROC curve for the custom CNN. The transfer model achieved a higher study AUC (0.8527) as reported in Table 2.

5. Practical challenges and fixes

Dataset path issue

The initial MURA runs failed because a placeholder path was used. The problem was solved by downloading and extracting MURA into the working folder and passing the real data directory.

CSV/path handling

The first script version was too rigid about CSV discovery. The loader was updated so that it could discover the dataset files more robustly.

Unreadable image files

Training initially failed because some images could not be decoded. The data-loading pipeline was changed to verify files with PIL and skip unreadable images. In the final run, 4 bad images were skipped.

Model behavior differences

The Part 1 experiments showed that small architectural or training choices matter. On Fashion-MNIST the MLP unexpectedly beat the CNN, while on CIFAR-10 the CNN was clearly stronger. This was reported honestly instead of being hidden.

6. Conclusion

This assignment showed that architecture choice must match the task. On CIFAR-10, the CNN was far better than the MLP because convolution exploits spatial structure. On Fashion-MNIST, the current MLP outperformed the small CNN in this specific run, which suggests that the CNN needed better tuning or more capacity. On MURA, transfer learning gave the strongest result by a clear margin and was the best overall solution for the X-ray task.

The final best models were the Fashion-MNIST MLP for grayscale clothing classification, the CIFAR-10 CNN for natural image classification, and the EfficientNetB0 transfer-learning model for study-level abnormality detection on MURA.

Appendix A - Execution evidence

The following screenshots were captured from the terminal after the final MURA runs. They show that both models trained successfully and saved their outputs.

```
(base) georgestamatogiannopoulos@Georges-MacBook-Pro deep_learning_assignment_simple % cd /Users/georgestamatogiannopoulos/Desktop/Deep_Learning/deep_learning_assignment_simple
rm -rf outputs/mura_custom_cnn
rm -rf outputs/mura_transfer
python part2_mura_custom_cnn.py --data_dir "/Users/georgestamatogiannopoulos/Desktop/Deep_Learning/deep_learning_assignment_simple/MURA-v1.1"
Loading MURA...
Skipped bad images: 4
Skipped bad images: 0
Train images: 36808
Valid images: 3197
Class weight: {0: 0.8399340770791075, 1: 1.2354367121652867}
Epoch 1/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 263s 127ms/step - accuracy: 0.5121 - auc: 0.5364 - loss: 0.6902 - val_accuracy: 0.5678 - val_auc: 0.6054 - val_loss: 0.6694 - learning_rate: 0.0010
Epoch 2/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 264s 127ms/step - accuracy: 0.5461 - auc: 0.5949 - loss: 0.6754 - val_accuracy: 0.5958 - val_auc: 0.6461 - val_loss: 0.6517 - learning_rate: 0.0010
Epoch 3/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 267s 129ms/step - accuracy: 0.5804 - auc: 0.6407 - loss: 0.6576 - val_accuracy: 0.6115 - val_auc: 0.6902 - val_loss: 0.6411 - learning_rate: 0.0010
Epoch 4/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 281s 136ms/step - accuracy: 0.6193 - auc: 0.6828 - loss: 0.6362 - val_accuracy: 0.6525 - val_auc: 0.7182 - val_loss: 0.6111 - learning_rate: 0.0010
Epoch 5/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 283s 136ms/step - accuracy: 0.6414 - auc: 0.7073 - loss: 0.6191 - val_accuracy: 0.6474 - val_auc: 0.7250 - val_loss: 0.6135 - learning_rate: 0.0010
Epoch 6/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 282s 136ms/step - accuracy: 0.6613 - auc: 0.7258 - loss: 0.6057 - val_accuracy: 0.6669 - val_auc: 0.7486 - val_loss: 0.6003 - learning_rate: 0.0010
Epoch 7/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 283s 137ms/step - accuracy: 0.6830 - auc: 0.7489 - loss: 0.5884 - val_accuracy: 0.7096 - val_auc: 0.7605 - val_loss: 0.5647 - learning_rate: 0.0010
Epoch 8/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 282s 136ms/step - accuracy: 0.6946 - auc: 0.7602 - loss: 0.5784 - val_accuracy: 0.7172 - val_auc: 0.7654 - val_loss: 0.5556 - learning_rate: 0.0010
{'study_auc': 0.7616121793609998, 'study_accuracy': 0.6914095079232694, 'study_kappa': 0.3550633408251025, 'valid_studies': 1199}
Done. Files saved in outputs/mura_custom_cnn
```

Appendix Figure A1. Final custom CNN training run on MURA.

```
(base) georgestamatogiannopoulos@Georges-MacBook-Pro deep_learning_assignment_simple % python part2_mura_transfer.py --data_dir "/Users/georgestamatogiannopoulos/Desktop/Deep_Learning/deep_learning_assignment_simple/MURA-v1.1"
Loading MURA...
Skipped bad images: 4
Skipped bad images: 0
Train images: 36808
Valid images: 3197
Class weight: {0: 0.8399340770791075, 1: 1.2354367121652867}
Downloading data from https://storage.googleapis.com/keras-applications/efficientnetb0_notop.h5
16705208/16705208 ━━━━━━━━━━━━━━━━━━━━ 3s 0us/step
Training classification head...
Epoch 1/4
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 183s 87ms/step - accuracy: 0.6253 - auc: 0.6765 - loss: 0.6416 - val_accuracy: 0.7167 - val_auc: 0.7894 - val_loss: 0.5561 - learning_rate: 0.0010
Epoch 2/4
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 205s 99ms/step - accuracy: 0.6849 - auc: 0.7541 - loss: 0.5865 - val_accuracy: 0.7465 - val_auc: 0.8059 - val_loss: 0.5233 - learning_rate: 0.0010
Epoch 3/4
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 197s 95ms/step - accuracy: 0.6933 - auc: 0.7609 - loss: 0.5816 - val_accuracy: 0.7381 - val_auc: 0.8063 - val_loss: 0.5241 - learning_rate: 0.0010
Epoch 4/4
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 210s 101ms/step - accuracy: 0.6968 - auc: 0.7648 - loss: 0.5765 - val_accuracy: 0.7378 - val_auc: 0.7997 - val_loss: 0.5298 - learning_rate: 0.0010
Fine-tuning top layers...
Epoch 5/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 243s 116ms/step - accuracy: 0.6571 - auc: 0.7213 - loss: 0.6221 - val_accuracy: 0.7503 - val_auc: 0.8083 - val_loss: 0.5180 - learning_rate: 1.0000e-05
Epoch 6/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 239s 115ms/step - accuracy: 0.7009 - auc: 0.7691 - loss: 0.5741 - val_accuracy: 0.7626 - val_auc: 0.8229 - val_loss: 0.5003 - learning_rate: 1.0000e-05
Epoch 7/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 246s 118ms/step - accuracy: 0.7189 - auc: 0.7907 - loss: 0.5520 - val_accuracy: 0.7715 - val_auc: 0.8321 - val_loss: 0.4884 - learning_rate: 1.0000e-05
Epoch 8/8
2071/2071 ━━━━━━━━━━━━━━━━━━━━ 232s 112ms/step - accuracy: 0.7232 - auc: 0.7962 - loss: 0.5445 - val_accuracy: 0.7759 - val_auc: 0.8373 - val_loss: 0.4816 - learning_rate: 1.0000e-05
{'study_auc': 0.852718366336912, 'study_accuracy': 0.7848206839032527, 'study_kappa': 0.5547459964246337, 'valid_studies': 1199}
Done. Files saved in outputs/mura_transfer
(base) georgestamatogiannopoulos@Georges-MacBook-Pro deep_learning_assignment_simple %
```

Appendix Figure A2. Final EfficientNetB0 transfer-learning run on MURA.